



MicroRobotics-Final

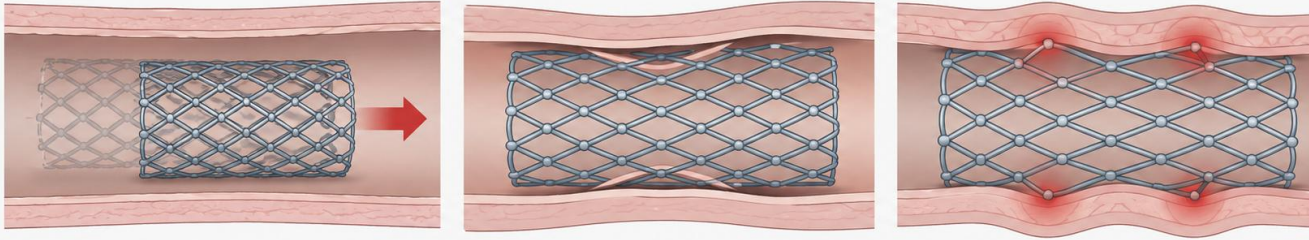
Magnetically Triggered Bistable Kirigami Micro-Scaffold for Temporary Support in Confined Soft Lumens

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Current devices can open a lumen, but they still struggle to provide **controlled, localized, and stable support** after deployment.

[1] Brinkmann et al., *Gastroenterology Report*, 2025.

[2] FDA, *Non-Clinical Engineering Tests and Recommended Labeling for Intravascular Stents*, 2010.

Migration

Stent displacement can reopen the target region or cause obstruction. Reported migration rates for fully covered biliary metal stents range from 5%–40%. [1]

Poor Fit

Insufficient radial outward force can leave gaps between the device and the lumen wall, reducing support stability and wall contact. [2]

Pressure Injury

Excessive radial outward force can over-compress soft tissue and increase the risk of local damage. [2]

Controlled Deployment

The device should avoid premature expansion during transport and switch state only when triggered.

Stable Support

After deployment, the structure should resist displacement and provide reliable support.

Safe Wall Contact

The outward force should be sufficient for contact but low enough to reduce tissue damage risk.

Key Challenge

Not simply generating force, but controlling **when, where, and how** that force is applied.

Design Idea: Separate the Functions



Bistable Kirigami hairpin

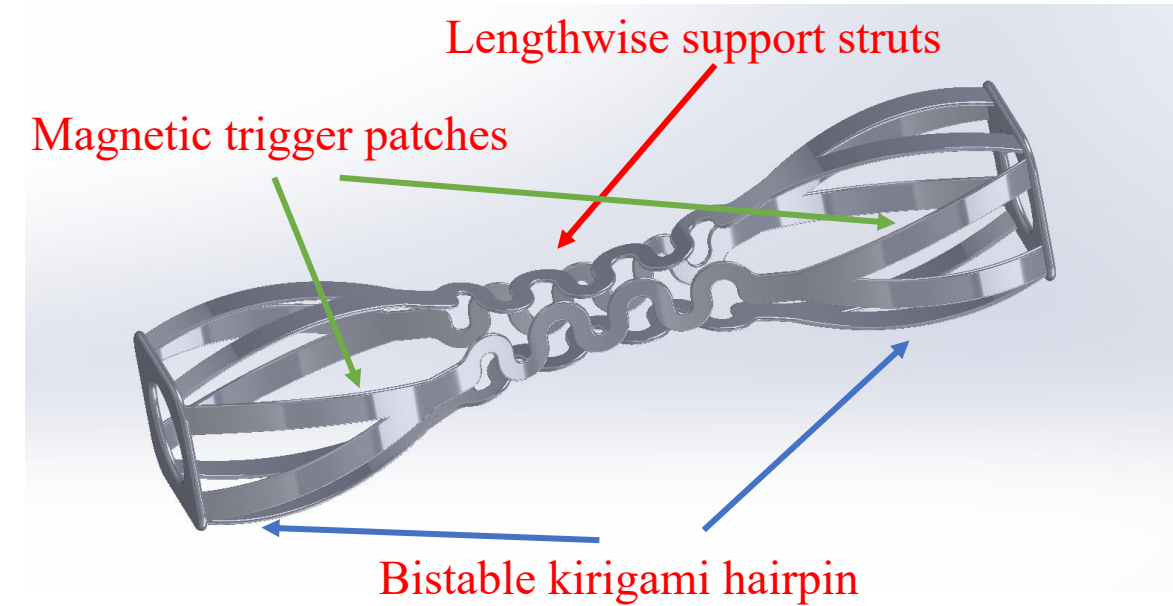
Enable large transformation and snap-through state switching.

Lengthwise support struts

Form the load-bearing scaffold for local lumen support and anti-migration.

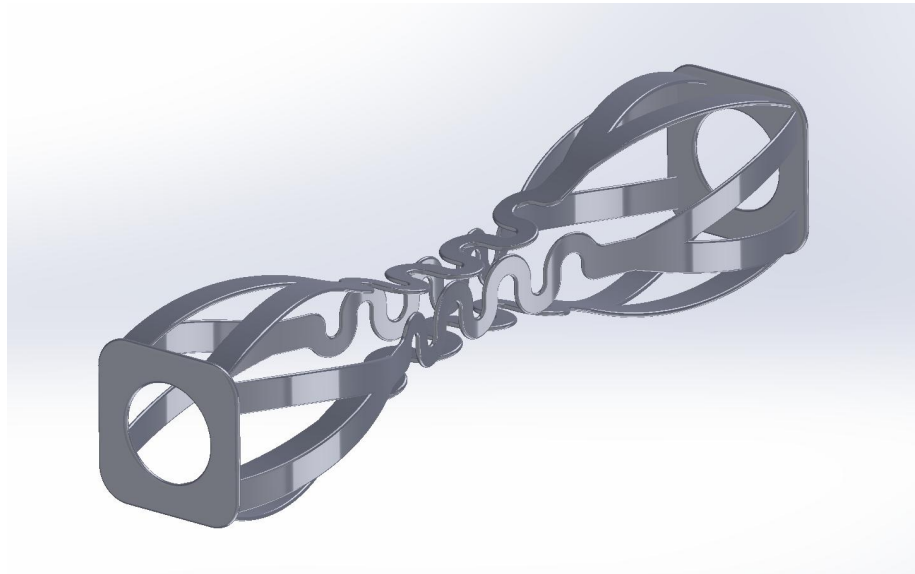
Magnetic trigger patches

Convert external magnetic input into local triggering torque.



Functional module map

Two-State Operation



Compressed Delivery
State

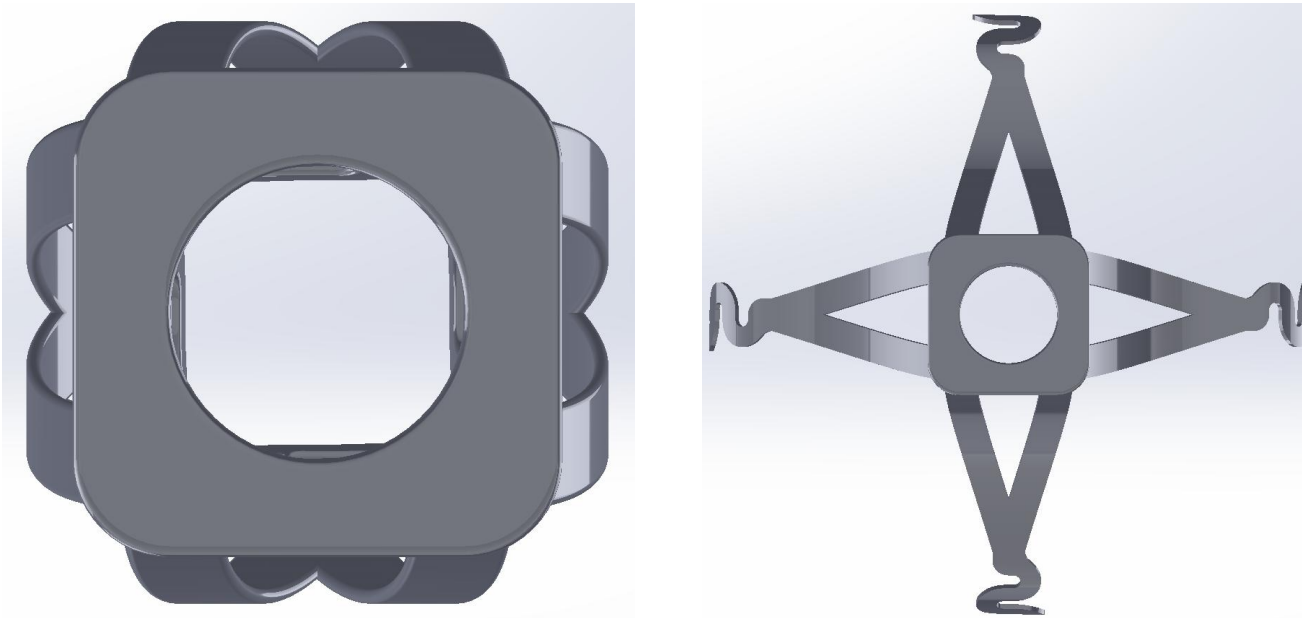
Magnetic Triggering
Induces snap-through



Deployed Support State

- **Low-profile geometry** for transport through confined lumens
- **End-rings remain locked** in the pre-deployment configuration
- **Kirigami end-rings snap open** into the deployed state
- **Lengthwise struts expand outward** to form a support cage

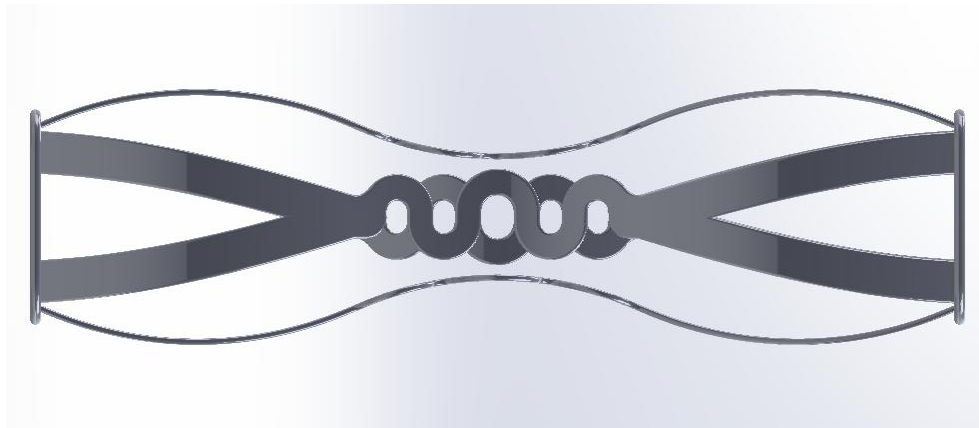
Parameter Introduction



Maximum diameter on the side

2.67mm → **7.46mm**

Expansion ratio **280%**



One-piece integrated body

Total length **8.7mm**

Patch thickness **0.02mm**

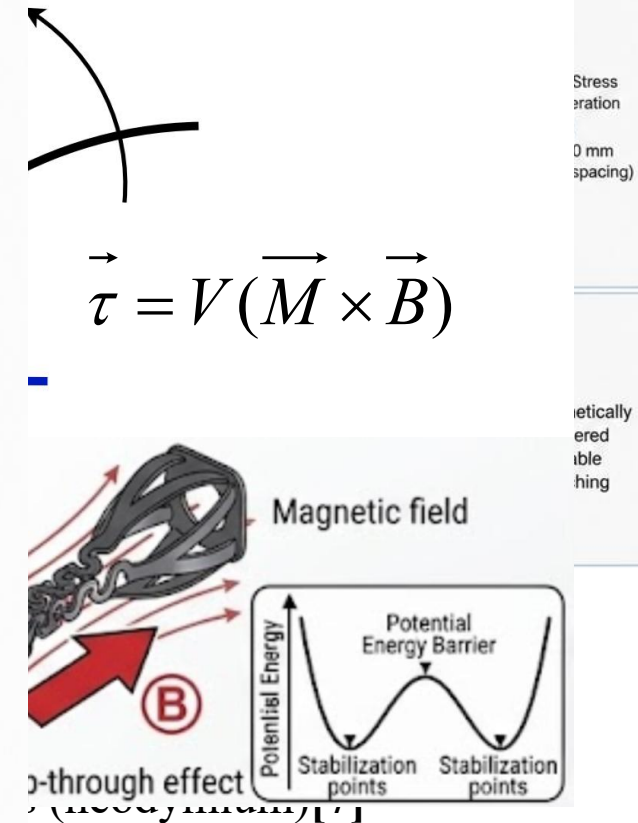
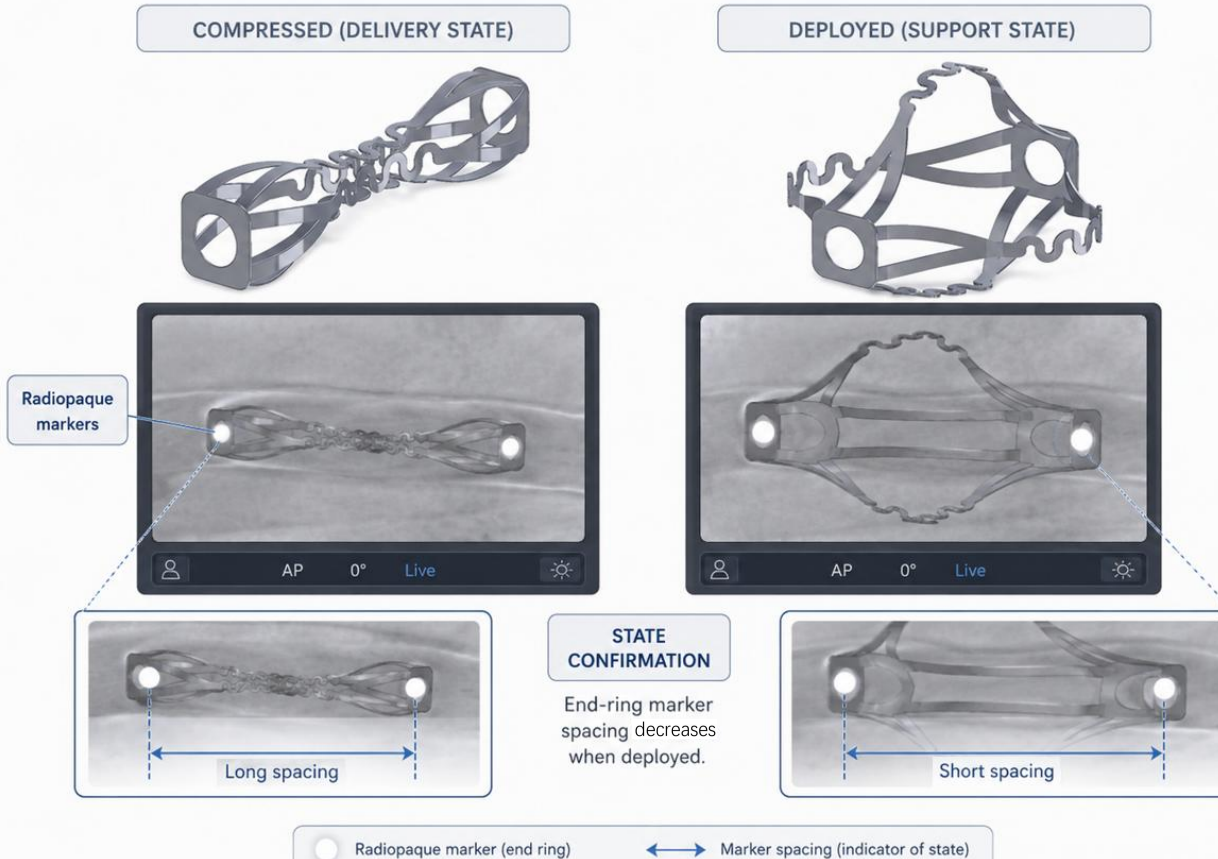
System Implementation Concept



TRACKING & STATE CONFIRMATION

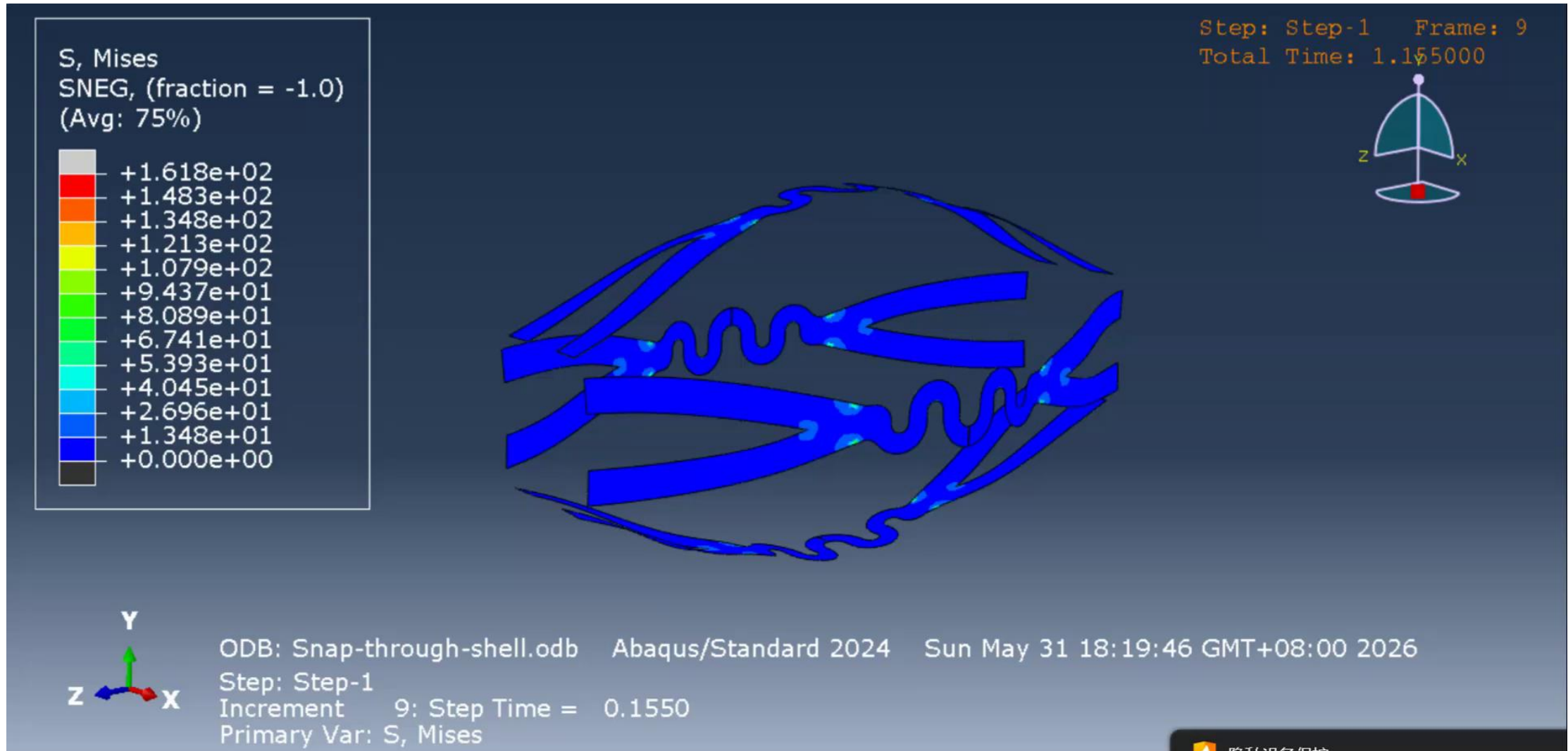
COMPRESSED (DELIVERY STATE)

DEPLOYED (SUPPORT STATE)



Deploy at target site: compressed delivery + magnetic state switching [6]
 Confirm position & state: radiopaque markers indicate position and deployment state [6, 8]

FEA Simulation Result



What this project addresses

Controlled Deployment

Target-site magnetic triggering enables on-demand state switching.

Stable Support

Deployed struts form a local support cage.

Function Separation

Actuation, locking, and support are assigned to different modules.

What remains future work

Clinical Validation

Not tested *in vivo*.

Navigation Control

Magnetic actuation is not used for autonomous vascular navigation.

Tissue Interaction

Realistic tissue mechanics, fatigue, and biocompatibility remain untested.

Prototype Fabrication

PI shell fabrication and magnetic coating require experimental validation.

This project validates a mechanical microrobotic concept, not a clinically ready device.

Problem-driven design:

The project targets post-deployment failures in Lumens: migration, poor fit, and pressure injury. [1,3]

Modular microrobotic architecture:

Kirigami end-rings, magnetic patches, and Lengthwise struts separate triggering, locking, and support. [4,5,6]

CAD–FEA validation pathway:

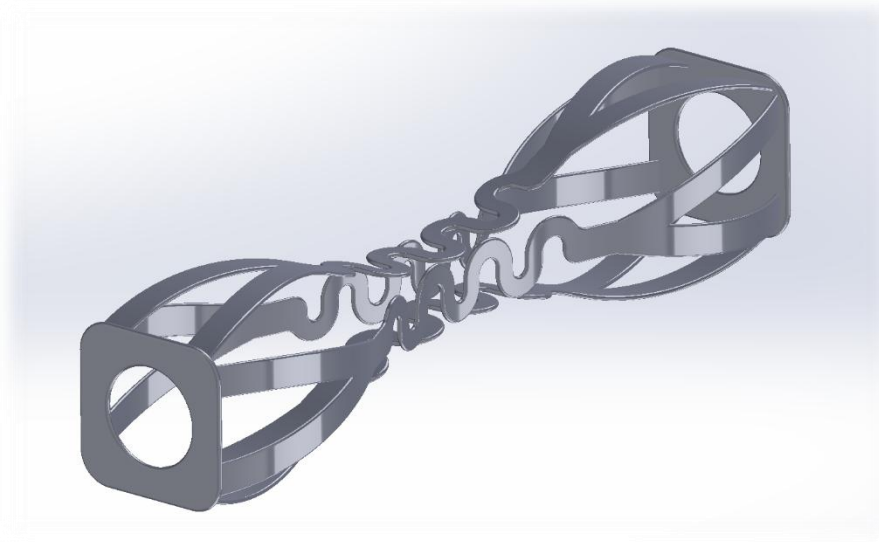
Geometry design and finite element analysis are used to evaluate snap-through, support force, and stress response. [3,5]

- [1] Chun, J. W. et al. Prevention of stent migration of covered self-expandable metal stents. *Gastroenterology Report*, 2025.
- [2] Babaei, S. et al. Kirigami-inspired stents for sustained local delivery of therapeutics. *Nature Materials*, 2021.
- [3] U.S. FDA. *Non-Clinical Engineering Tests and Recommended Labeling for Intravascular Stents and Associated Delivery Systems*, 2010.
- [4] Kim, T. et al. Shape transformable bifurcated stents. *Scientific Reports*, 2018.
- [5] Yang, Y., Dias, M. A., Holmes, D. P. Multistable Kirigami for Tunable Architected Materials. *Physical Review Materials*, 2018.
- [6] Zhou, J. et al. System integration of magnetic medical microrobots. *Frontiers in Robotics and AI*, 2023.
- [7] Dong, Z. et al. Microfabrication of functional polyimide films and microstructures for flexible MEMS applications. *Microsystems & Nanoengineering*, 2023.
- [8] Boston Scientific / FDA. *Innova Vascular Self-Expanding Stent System Instructions / FDA PMA device document*.



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THANK YOU



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